

Ibrahim Tayeb

US Citizen | 510-298-7759 | ibrahimtayeb30@gmail.com | [linkedin.com/in/ibrahimtayeb](https://www.linkedin.com/in/ibrahimtayeb) | github.com/IbrahimTayeb

EDUCATION

University of Illinois - Urbana-Champaign

Urbana, IL

Bachelor of Science in Computer Science

Expected Graduation: May 2027

- GPA: 3.7/4.0
- Coursework: Computer Architecture, Data Structures, Algorithms, System Programming, Software Design, Applied Machine Learning, Artificial Intelligence, Computer Security, Distributed Systems, Probability & Statistics for Computer Science, Database Systems, Numerical Methods, Cryptography, Linear Algebra, Discrete Structures

EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Uber

San Francisco, CA

- Designed & built order auto-approval system automating earner payout resolution contributing to \$11M in savings
- Architected 3-layer event-driven pipeline (Watcher → Scheduler → Executor) in Go, consuming real-time Kafka events, scheduling timers, and executing downstream approval actions with 99% execution success
- Built end-to-end testing infrastructure using synthetic event generation to validate the full pipeline
- Leveraged multi-modal LLMs and agentic AI to accelerate codebase exploration, debugging, and development

Robotics Software Engineer

August 2023 – May 2026

University of Illinois - Urbana-Champaign (NASA Lunabotics)

Urbana, IL

- Developed algorithms in C++/Python for autonomous navigation, leveraging ROS2 & SLAM toolbox to enable real-time path planning, resulting in a 25% improvement in movement efficiency and reduced collision incidents
- Integrated LIDAR sensors, computer vision cameras, and motor control units with Raspberry Pi microcomputers, achieving enhanced spatial awareness and precision control, improving positioning accuracy by 30%
- Conducted iterative testing and troubleshooting, utilizing simulations with Gazebo, reducing system errors by 20%

Software Engineer Intern

May 2025 – Aug. 2025

Amway

Grand Rapids, MI

- Architected scalable AWS infrastructure, containerizing applications with Docker & deploying to ECS Fargate
- Automated CI/CD pipelines using CodePipeline, achieving 99.9% uptime and reducing deployment time by 75%
- Built Streamlit dashboard with API integration, session state management, & interactive UI for data visualization
- Designed modular service architecture consuming REST APIs with OAuth2 authentication, token caching, error handling, and data transformation layers for seamless backend integration, improving system reliability by 60%

Software Engineer Intern

May 2024 – Aug. 2024

Amway

Grand Rapids, MI

- Designed and developed an internal full-stack web application using ASP.NET, HTML, CSS, JavaScript (frontend), and C#, SQL Server (backend) to streamline the process of retrieving shipment pallet stacking schematics
- Leveraged MS SQL Server to handle data storage and retrieval, allowing engineers to efficiently filter results based on specific criteria, improving decision-making and reducing shipment preparation time by 30%

PROJECTS

Developer LinkTree | *TypeScript, React, Node.js, Express, Prisma, SQLite, JWT, Tailwind CSS*

- Built RESTful API with Node.js & Express, integrated Prisma ORM & SQLite for data modeling and persistence
- Implemented full user authentication using JWT, including signup, login, logout, and token expiration
- Developed a responsive frontend with protected routes, dynamic forms, and dashboard UI

TECHNICAL SKILLS

Languages: Python, C++, Java, C#, JavaScript, HTML/CSS, SQL

Frameworks: React, Express, Angular, Node.js, Flask, AWS cdk, .NET, ROS2

Developer Tools: Git, GitHub, AWS, Docker/Podman, Bitbucket, MongoDB, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, MS SQL Server, Linux (Ubuntu 22.04), Postman, Jupyter Notebook, Elasticsearch

Libraries: Pandas, NumPy, Matplotlib, NLTK, Scikit-Learn, TensorFlow, PyTorch, Streamlit, Requests, Plotly